STAT 410
Fall 2017

Name ANSWERS

Version B

NetID _____

| Page | Earned |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| Total | / 100 |

| 1 | 2 | 3 |
|---|---|---|
| MWF 9:00 | TuTh 12:30 | TuTh 2:00 |

*Please circle your section.*

# Exam 2

*Be sure to show all your work; your partial credit might depend on it.*

*Put your final answers at the end of your work, and mark them clearly.*

*No credit will be given without supporting work.*

The exam is closed book and closed notes. You are allowed to use a calculator and one 8½" x 11" sheet with notes.

*Turn in all scratch paper with your exam.*

---

Academic Integrity

The University statement on your obligation to maintain academic integrity is:

If you engage in an act of academic dishonesty, you become liable to severe disciplinary action. Such acts include cheating; falsification or invention of information or citation in an academic endeavor; helping or attempting to help others commit academic infractions; plagiarism; offering bribes, favors, or threats; academic interference; computer related infractions; and failure to comply with research regulations.

Rule 33 of the Code of Policies and Regulations Applying to All Students gives complete details of rules governing academic integrity for all students. You are responsible for knowing and abiding by these rules.

**1.** Suppose that company A and company B are in the same industry sector, and the prices of their stocks, \$X per share for company A and \$Y per share for company B, vary from day to day randomly according to a bivariate normal distribution with parameters

$$\mu_X = 75, \quad \sigma_X = 9, \quad \mu_Y = 23, \quad \sigma_Y = 5, \quad \rho = 0.6.$$

a) (10) Suppose that on a given day the price of stock for company B is \$26. What is the probability that the price of stock for company A exceeds \$84? That is, find $P(X > 84 \mid Y = 26)$.

Given $Y = 26$, $X$ has Normal distribution

with mean $\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y) = 75 + 0.6 \cdot \frac{9}{5} \cdot (26 - 23) = 78.24$

and variance $\left(1 - \rho^2\right) \cdot \sigma_X^2 = \left(1 - 0.6^2\right) \cdot 9^2 = 51.84$

( standard deviation = $7.2$ ).

$$P(X > 84 \mid Y = 26) = P\left(Z > \frac{84 - 78.24}{7.2}\right) = P(Z > 0.80) = \mathbf{0.2119}.$$

b) (10) What is the probability that 1 share of company A stock is worth more than 3 shares of company B stock? That is, find $P(X > 3\,Y)$.

Want $P(X > 3\,Y) = P(X - 3\,Y > 0) = ?$

$X - 3\,Y$ has Normal distribution,

$E(X - 3\,Y) = \mu_X - 3\,\mu_Y = 75 - 3 \cdot 23 = 6,$

$\mathrm{Var}(X - 3\,Y) = \sigma_X^2 - 6\,\sigma_{XY} + 9\sigma_Y^2 = \sigma_X^2 - 6\,\rho\,\sigma_X\,\sigma_Y + 9\sigma_Y^2$

$= 9^2 - 6 \cdot 0.6 \cdot 9 \cdot 5 + 9 \cdot 5^2 = 144$ ( standard deviation = $12$ ).

$$P(X - 3\,Y > 0) = P\left(Z > \frac{0 - 6}{12}\right) = P(Z > -0.50) = \mathbf{0.6915}.$$

**2.** (8) The number of deer struck by cars on the highway between Anytown and Someville in November follows a Poisson process with the average rate of 2.5 deer per week. Find the probability that the eighth deer is struck during the fourth week in November.

$X_t$ = number of deer struck in $t$ weeks. Poisson$(\lambda t)$

$T_k$ = the time when the $k$th deer is struck. Gamma, $\alpha = k$.

2.5 deer per week $\Rightarrow$ $\lambda = 2.5$.

If $T_\alpha$ has a Gamma$(\alpha, \theta = 1/\lambda)$ distribution, where $\alpha$ is an integer, then

$$P(T_\alpha \le t) = P(X_t \ge \alpha) \quad \text{and} \quad P(T_\alpha > t) = P(X_t \le \alpha - 1),$$

where $X_t$ has a Poisson$(\lambda t)$ distribution.

$$P(3 < T_8 < 4) = P(T_8 > 3) - P(T_8 > 4) = P(X_3 \le 7) - P(X_4 \le 7)$$

$$= P(\text{Poisson}(7.5) \le 7) - P(\text{Poisson}(10.0) \le 7) = 0.525 - 0.220 = \mathbf{0.305}.$$

OR

$$P(3 < T_8 < 4) = \int_3^4 \frac{2.5^8}{\Gamma(8)} t^{8-1} e^{-2.5t}\, dt = \int_3^4 \frac{2.5^8}{7!} t^7 e^{-2.5t}\, dt = \ldots$$

**3.** Let $\delta > 0$. Let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with probability density function

$$f(x;\delta) = -\delta^2 \ln x \cdot x^{\delta-1}, \qquad 0 < x < 1, \qquad \text{zero otherwise.}$$

Note: Since $0 < x < 1$, $\ln x < 0$.

a) (4) Find a sufficient statistic $Y = u(X_1, X_2, \ldots, X_n)$ for $\delta$.

$\lambda > 0$ $\qquad f(x;\lambda) = -\lambda^2 \ln x \cdot x^{\lambda-1}, \qquad 0 < x < 1, \qquad$ zero otherwise.

$$\prod_{i=1}^{n} f(x_i;\lambda) = \prod_{i=1}^{n} \lambda^2 (-\ln x_i) \cdot x_i^{\lambda-1} = \lambda^{2n} \prod_{i=1}^{n} (-\ln x_i) \cdot \left( \prod_{i=1}^{n} x_i \right)^{\lambda-1}.$$

By Factorization Theorem, $\prod_{i=1}^{n} X_i$ is a sufficient statistic for $\lambda$.

OR

$f(x;\lambda) = \exp\{(\lambda-1)\ln x + 2\ln\lambda + \ln(-\ln x)\}$. $\qquad K(x) = \ln x$.

$\Rightarrow \quad \sum_{i=1}^{n} K(X_i) = \sum_{i=1}^{n} \ln X_i$ is a sufficient statistic for $\lambda$.

$\Rightarrow \quad -\sum_{i=1}^{n} \ln X_i$ is a sufficient statistic for $\lambda$.

**3.** (continued) $f(x;\delta) = -\delta^2 \ln x \cdot x^{\delta-1}, \qquad 0 < x < 1.$

b) (16) (i) Construct a consistent estimator of $\delta$.

(ii) Prove that your estimator is indeed a consistent estimator of $\delta$.

$\lambda > 0 \qquad f(x;\lambda) = -\lambda^2 \ln x \cdot x^{\lambda-1}, \qquad 0 < x < 1, \qquad$ zero otherwise.

(i) $1 = \int_0^1 \left(-\lambda^2 \ln x \cdot x^{\lambda-1}\right) dx. \qquad \Rightarrow \qquad \int_0^1 \left(-\ln x \cdot x^{\lambda-1}\right) dx = \frac{1}{\lambda^2}, \qquad \lambda > 0.$

$$\mathrm{E}(\mathrm{X}^k) = \int_0^1 x^k \cdot \left(-\lambda^2 \ln x \cdot x^{\lambda-1}\right) dx = \lambda^2 \int_0^1 \left(-\ln x \cdot x^{\lambda+k-1}\right) dx = \frac{\lambda^2}{(\lambda+k)^2},$$

$k > -\lambda.$

$$\mu = \mathrm{E}(\mathrm{X}) = \mathrm{E}(\mathrm{X}^1) = \frac{\lambda^2}{(\lambda+1)^2}.$$

OR $\qquad \mu = \mathrm{E}(\mathrm{X}) = \int_0^1 x \cdot \left(-\lambda^2 \ln x \cdot x^{\lambda-1}\right) dx = \ldots$ by parts $\ldots$

$$\overline{\mathrm{X}} = \frac{\tilde{\lambda}^2}{(\tilde{\lambda}+1)^2}. \qquad \Rightarrow \qquad \sqrt{\overline{\mathrm{X}}} = \frac{\tilde{\lambda}}{\tilde{\lambda}+1}. \qquad \Rightarrow \qquad \tilde{\lambda} = \frac{\sqrt{\overline{\mathrm{X}}}}{1-\sqrt{\overline{\mathrm{X}}}}.$$

(ii) By WLLN, $\qquad \overline{\mathrm{X}} \xrightarrow{P} \mu = \frac{\lambda^2}{(\lambda+1)^2}.$

Since $g(x) = \frac{\sqrt{x}}{1-\sqrt{x}}$ is continuous at $\mu$,

$$\tilde{\lambda} = g(\overline{\mathrm{X}}) \xrightarrow{P} g(\mu) = \frac{\frac{\lambda}{\lambda+1}}{1-\frac{\lambda}{\lambda+1}} = \lambda.$$

$\tilde{\lambda}$ is a consistent estimator of $\lambda$.

**OR**

(i) $\mathrm{L}(\lambda) = \prod_{i=1}^{n} f(x_i;\lambda) = \prod_{i=1}^{n} \lambda^2 (-\ln x_i)\cdot x_i^{\lambda-1} = \lambda^{2n} \prod_{i=1}^{n} (-\ln x_i)\cdot \left(\prod_{i=1}^{n} x_i\right)^{\lambda-1}$.

$$\ln \mathrm{L}(\lambda) = 2n \ln \lambda + \sum_{i=1}^{n} \ln(-\ln x_i) + (\lambda - 1)\sum_{i=1}^{n} \ln x_i .$$

$$\frac{d}{d\lambda} \ln \mathrm{L}(\lambda) = \frac{2n}{\lambda} + \sum_{i=1}^{n} \ln x_i = 0. \qquad \Rightarrow \qquad \hat{\lambda} = -\frac{2n}{\sum_{i=1}^{n} \ln \mathrm{X}_i}.$$

(ii) Let $\mathrm{W} = -\ln \mathrm{X}$.

$$x = e^{-w} \qquad \frac{dx}{dw} = -e^{-w}$$

$$f_{\mathrm{W}}(w) = f_{\mathrm{X}}(e^{-w})\cdot \left|\frac{dx}{dw}\right| = \lambda^2 w e^{-(\lambda-1)w}\cdot e^{-w} = \frac{\lambda^2}{\Gamma(2)} w^{2-1} e^{-\lambda w}, \quad w > 0.$$

$\Rightarrow$ $\mathrm{W} = -\ln \mathrm{X}$ has a $\mathrm{Gamma}\left(\alpha = 2, \theta = \frac{1}{\lambda}\right)$ distribution.

By WLLN, $\overline{\mathrm{W}} \xrightarrow{P} \mu_{\mathrm{W}} = \alpha\theta = \frac{2}{\lambda}$.

OR $\mu_{\mathrm{W}} = \mathrm{E}(-\ln \mathrm{X}) = \int_0^1 \lambda^2 (\ln x)^2 \cdot x^{\lambda-1}\, dx$ = … by parts … twice …

Since $g(x) = \frac{2}{x}$ is continuous at $\mu_{\mathrm{W}}$,

$$\hat{\lambda} = \frac{2n}{\sum_{i=1}^{n} \mathrm{W}_i} = \frac{2}{\overline{\mathrm{W}}} = g(\overline{\mathrm{W}}) \xrightarrow{P} g(\mu_{\mathrm{W}}) = g\left(\frac{2}{\lambda}\right) = \lambda.$$

$\hat{\lambda}$ is a consistent estimator of $\lambda$.

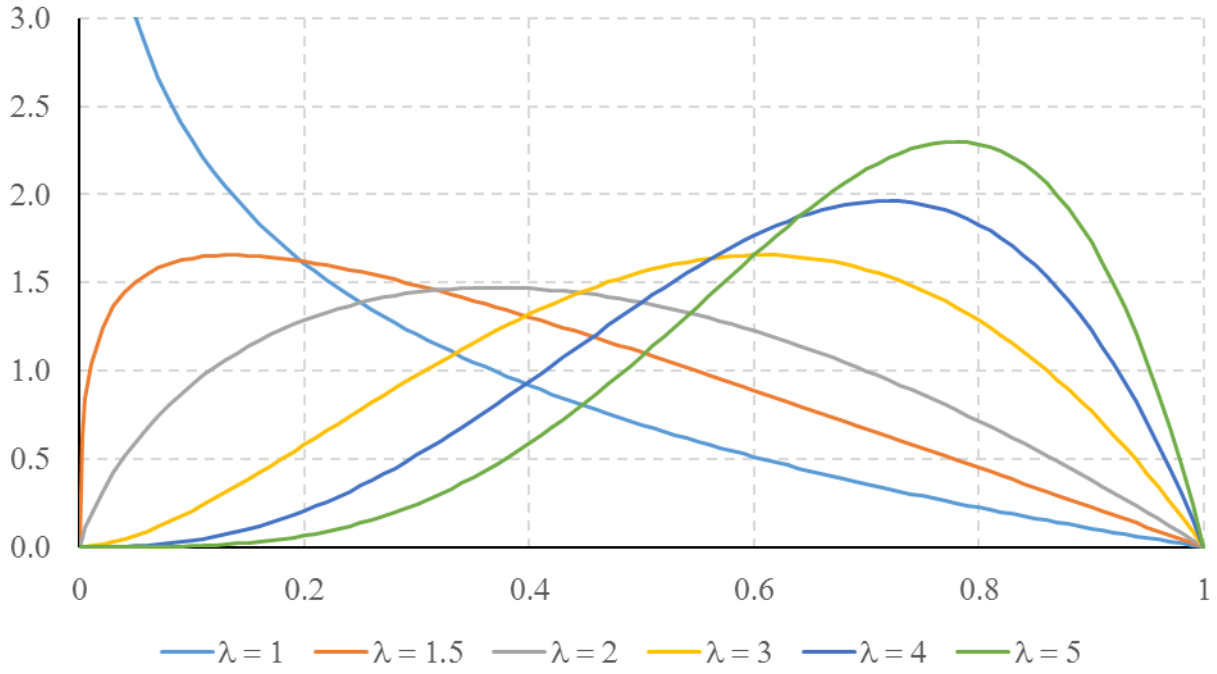
**OR**

(i) $$\mathrm{E}\left(\frac{1}{-\ln \mathrm{X}}\right) = \int_0^1 \frac{1}{-\ln x} \cdot \left(-\lambda^2 \ln x \cdot x^{\lambda-1}\right) dx = \int_0^1 \lambda^2 x^{\lambda-1}\, dx = \lambda.$$

Consider $$\hat{\hat{\lambda}} = \overline{\frac{1}{-\ln \mathrm{X}}} = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{-\ln \mathrm{X}_i} = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{\ln \frac{1}{\mathrm{X}_i}}.$$

(ii) By WLLN, $$\hat{\hat{\lambda}} = \overline{\frac{1}{-\ln \mathrm{X}}} \xrightarrow{P} \mathrm{E}\left(\frac{1}{-\ln \mathrm{X}}\right) = \lambda.$$

$\hat{\hat{\lambda}}$ is a consistent estimator of $\lambda$.



**4.** Let $\tau \in \mathbf{R}$. Let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with probability density function

$$f(x;\tau) = 3e^{-3x+3\tau}, \qquad x > \tau, \qquad \text{zero otherwise.}$$

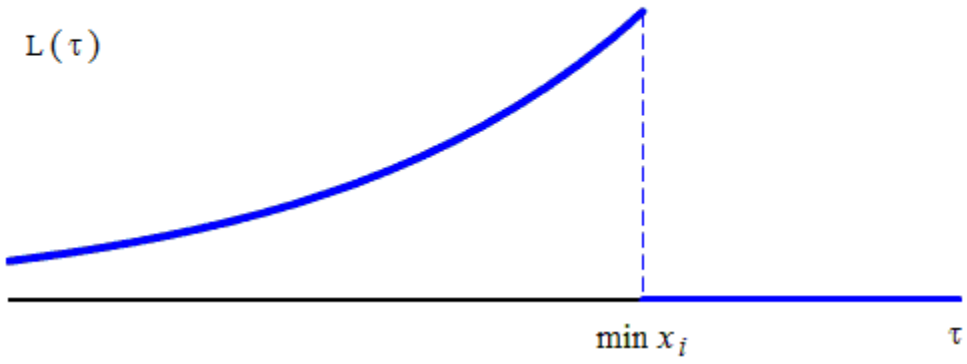
a) (10) (i) Obtain the maximum likelihood estimator of $\tau$, $\hat{\tau}$.

(ii) Suppose $n = 5$, and $x_1 = 2.20$, $x_2 = 2.85$, $x_3 = 2.30$, $x_4 = 2.15$, $x_5 = 2.50$. Obtain the maximum likelihood estimate of $\tau$, $\hat{\tau}$.

(i) $$L(\tau) = \prod_{i=1}^{n} f(x_i;\tau) = \begin{cases} \prod_{i=1}^{n} 3e^{-3x_i+3\tau} & x_1 > \tau, x_2 > \tau, \ldots, x_n > \tau \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} 3^n e^{-3\Sigma x_i + 3n\tau} & \min x_i > \tau \\ 0 & \tau > \min x_i \end{cases}$$

$$\ln L(\tau) = n \ln 3 - 3\sum_{i=1}^{n} x_i + 3n\tau. \qquad \frac{d}{d\tau} \ln L(\tau) = 3n = 0 \quad ???$$




$\Rightarrow \quad \hat{\tau} = \min X_i.$

(ii) $\hat{\tau} = \min x_i = \mathbf{2.15}.$

**4.** (continued) $f(x;\tau) = 3e^{-3x+3\tau}, \qquad x > \tau.$

b) (10) Is the maximum likelihood estimator $\hat{\tau}$ an unbiased estimator of $\tau$?

If $\hat{\tau}$ is not an unbiased estimator of $\tau$, construct an unbiased estimator of $\tau$ based on $\hat{\tau}$.

$$F_X(x) = \int_{\tau}^{x} 3e^{-3u+3\tau}\,du = 1 - e^{-3x+3\tau} = 1 - e^{-3(x-\tau)}, \qquad x > \tau.$$

$$F_{\min X_i}(x) = 1 - (1 - F_X(x))^n = 1 - e^{-3n(x-\tau)}, \qquad x > \tau.$$

$$f_{\min X_i}(x) = 3ne^{-3n(x-\tau)}, \qquad x > \tau.$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0. \qquad \Rightarrow \qquad E(Y) = \frac{1}{\lambda}.$$

$$f_Y(y) = \lambda e^{-\lambda(y-\theta)}, \quad y > \theta. \qquad \Rightarrow \qquad E(Y) = \theta + \frac{1}{\lambda}.$$

$$\Rightarrow \qquad E(\min X_i) = \tau + \frac{1}{3n}.$$

OR $\qquad E(\min X_i) = \int_{\delta}^{\infty} x \cdot 3ne^{-3n(x-\tau)}\,dx = \ldots$ by parts …

$E(\min X_i) = \tau + \frac{1}{3n} \neq \tau.$ $\qquad \hat{\tau}$ is NOT an unbiased estimator of $\tau$.

Consider $\quad \hat{\hat{\tau}} = \min X_i - \frac{1}{3n}.$

$E(\hat{\hat{\tau}}) = \tau.$ $\qquad \hat{\hat{\tau}}$ is an unbiased estimator of $\tau$.

**5.** Let $\beta > 0$. Let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with probability density function

$$f(x;\beta) = \beta e^{-\beta x + 2\beta}, \qquad x > 2, \qquad \text{zero otherwise.}$$

a) (10) (i) Obtain the maximum likelihood estimator of $\beta$, $\hat{\beta}$.

(ii) Suppose $n = 5$, and $x_1 = 2.20$, $x_2 = 2.85$, $x_3 = 2.30$, $x_4 = 2.15$, $x_5 = 2.50$. Obtain the maximum likelihood estimate of $\beta$, $\hat{\beta}$.

(i) $$L(\beta) = \prod_{i=1}^{n} f(x_i;\beta) = \begin{cases} \prod_{i=1}^{n} \beta e^{-\beta x_i + 2\beta} & x_1 > 2, x_2 > 2, \ldots, x_n > 2 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \beta^n e^{-\beta \Sigma x_i + 2n\beta} & \min x_i > 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\ln L(\beta) = n \ln \beta - \beta \sum_{i=1}^{n} x_i + 2n\beta.$$

$$\frac{d}{d\beta} \ln L(\beta) = \frac{n}{\beta} - \sum_{i=1}^{n} x_i + 2n = 0.$$

$$\Rightarrow \qquad \hat{\beta} = \frac{n}{\sum_{i=1}^{n} X_i - 2n} = \frac{n}{\sum_{i=1}^{n} (X_i - 2)} = \frac{1}{\overline{X} - 2}.$$

(ii) $$\sum_{i=1}^{n} x_i = 12. \qquad \sum_{i=1}^{n} (x_i - 2) = 2. \qquad \bar{x} = 2.4.$$

$$\hat{\beta} = \frac{5}{2} = \mathbf{2.5}.$$

**5.** (continued) $f(x;\beta) = \beta e^{-\beta x + 2\beta}, \qquad x > 2.$

b) (10) Is the maximum likelihood estimator $\hat{\beta}$ an unbiased estimator of $\beta$?

If $\hat{\beta}$ is not an unbiased estimator of $\beta$, construct an unbiased estimator of $\beta$ based on $\hat{\beta}$.

"Hint": $W = X - 2$.

$$f_X(x) = \beta e^{-\beta x + 2\beta} = \beta e^{-\beta(x-2)}, \qquad x > 2.$$

$W = X - 2 \qquad x = w + 2 \qquad \frac{dx}{dw} = 1$

$$f_W(w) = \beta e^{-\beta w}, \qquad w > 0.$$

$W = X - 2$ has an $\text{Exponential}(\theta = \frac{1}{\beta}) = \text{Gamma}(\alpha = 1, \theta = \frac{1}{\beta})$ distribution.

$\Rightarrow \quad Y = \sum_{i=1}^{n} W_i$ has a $\text{Gamma}(\alpha = n, \theta = \frac{1}{\beta})$ distribution.

$$\hat{\beta} = \frac{n}{\sum_{i=1}^{n}(X_i - 2)} = \frac{n}{\sum_{i=1}^{n} W_i} = \frac{n}{Y}. \qquad E(Y^k) = \frac{\Gamma(n+k)}{\beta^k \Gamma(n)}.$$

$$E(\hat{\beta}) = E\left(\frac{n}{Y}\right) = n\,E\left(\frac{1}{Y}\right) = n\,E(Y^{-1}) = n \cdot \frac{\beta}{n-1} = \frac{n}{n-1} \cdot \beta \neq \beta.$$

$\Rightarrow \quad \hat{\beta}$ is NOT an unbiased estimator for $\beta$.

Consider $\hat{\hat{\beta}} = \frac{n-1}{n} \cdot \hat{\beta} = \frac{n-1}{\sum_{i=1}^{n} X_i - 2n} = \frac{n-1}{\sum_{i=1}^{n}(X_i - 2)}.$

Then $E(\hat{\hat{\beta}}) = \frac{n-1}{n} \cdot E(\hat{\beta}) = \beta.$ $\hat{\hat{\beta}}$ is an unbiased estimator for $\tau$.

**5.** (continued) $f(x;\beta) = \beta e^{-\beta x + 2\beta}, \qquad x > 2.$

c) (12) Suppose $n = 5,$ and $x_1 = 2.20,\ x_2 = 2.85,\ x_3 = 2.30,\ x_4 = 2.15,\ x_5 = 2.50.$
Construct a 95% confidence interval for $\beta$.

"Hint": $W = X - 2.$

$W = X - 2$ has an $\text{Exponential}(\theta = \frac{1}{\beta}) = \text{Gamma}(\alpha = 1, \theta = \frac{1}{\beta})$ distribution.

$\Rightarrow \quad Y = \sum_{i=1}^{n} W_i = \sum_{i=1}^{n} (X_i - 2)$ has a $\text{Gamma}(\alpha = n, \theta = \frac{1}{\beta})$ distribution.

$2Y/\theta = 2\beta \sum_{i=1}^{n} (X_i - 2)$ has a chi-square distribution with $r = 2\alpha = 2n$ d.f.

$\Rightarrow \quad P\left(\chi^2_{1-\alpha/2}(2n) < 2\beta \sum_{i=1}^{n} (X_i - 2) < \chi^2_{\alpha/2}(2n)\right) = 1 - \alpha.$

$\Rightarrow \quad P\left(\frac{\chi^2_{1-\alpha/2}(2n)}{2\sum_{i=1}^{n}(X_i - 2)} < \beta < \frac{\chi^2_{\alpha/2}(2n)}{2\sum_{i=1}^{n}(X_i - 2)}\right) = 1 - \alpha.$

A $(1-\alpha)100\ \%$ confidence interval for $\beta$: $\left(\frac{\chi^2_{1-\alpha/2}(2n)}{2\sum_{i=1}^{n}(X_i - 2)}, \frac{\chi^2_{\alpha/2}(2n)}{2\sum_{i=1}^{n}(X_i - 2)}\right).$

$\chi^2_{0.975}(10) = 3.247, \qquad \chi^2_{0.025}(10) = 20.48. \qquad \sum_{i=1}^{n}(x_i - 2) = 2.$

$\left(\frac{3.247}{2 \cdot 2}, \frac{20.48}{2 \cdot 2}\right)$ **(0.81175, 5.12)**